

Task 3:- what happens when you press the start button: —

It's time to boot up the system. The flow below shows the steps a computer system goes through before it shows you a working interface (in form of operating system).



Step 1:- Press the Power Button

When we press the power button on our computer system, a signal is sent to the PSU to allow power to flow. Imagine your body being powered off when we sleep. Once we wake up and receive oxygen, our body starts pumping blood and boots up.

Step 2:- Firmware Starts

Continuing our analogy from Step 1, once the body has started up, our core components are up and running, but our brain is not yet conscious. Like our bodies, a computer system contains firmware that allows all its components to start up. The central system that manages this is called Unified Extensible Firmware Interface (UEFI). Note:- we will often see the term BIOS mentioned instead of UEFI. BIOS does the same as UEFI, but has mainly been replaced by UEFI.

Step 3:- Power on Self Test

Now that our body is up and running, it is time to test if everything is functioning as it should. If something isn't, there will be some alarm signals. One of the routines that UEFI loads is power on self test, which tests if every required component is present, configured correctly.

Step 4:- Select Boot Device

Once our body is up and running, configured correctly and fully functional, our system searches for the location of our boot routine to start our consciousness. In our computer system, UEFI holds an ordered list which prioritizes on which device to look first for the boot up routine for the operating system.

Step 5:- Initiate Bootloader

Now our system knows part of our brain where our consciousness is located. At initiation

the "load routine" to start it. Our computer system follows a similar process: On selected boot device, the bootloader is initiated. This bootloader transfers operating system from the selected boot device to Random Access memory. Once OS is transferred, the UEFI gives control over different components to OS.

Quiz:-

- 1) What happens first when you press the power button?
 - A) The operating system starts loading
 - ☒ B) The PSU starts delivering power to components
 - C) The hard drive starts spinning
- 2) What is UEFI's role in boot process?
 - A) It runs your applications
 - ☒ B) It initializes and coordinates hardware components
 - C) It connects to the Internet
- 3) What does bootloader do?
 - A) Shuts down the computer safely
 - ☒ B) Loads operating system into RAM
 - C) Tests keyboard and mouse
- 3) Power on self Test (POST)
 - A) Whether your programs are up to date
 - ☒ B) Whether required hardware is present and working
 - C) Whether you're connected to Internet.
- 4) How does UEFI know which device to boot from?
 - A) It randomly selects a storage device
 - ☒ B) It follows a configured boot list priority
 - C) It always boots from newest device.

Overview:-

1. Press power button:- You press the power button, sending a signal to PSU to let power through. The system begins powering up.

2. Firmware starts (UEFI/BIOS):-

UEFI (or older BIOS) firmware initializes and coordinates components. It's the first software that runs.

3. Power on self Test (POST):- UEFI runs POST to check required hardware is present, configured, and functioning. Errors trigger beeps or alerts.

4. Select boot device:-

UEFI follows priority list to determine which device to boot from - typically SSD or HDD with OS installed.

5. Initiate Bootloader:- The bootloader loads operating system into RAM. Then UEFI hands control to OS, completing boot sequence.